

作业纸

课程名称: _____

班级: _____

教学班级: _____

姓名: _____

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$$2.4. y[n] = \sum_{k=-\infty}^{+\infty} x[k] h[n-k]$$

当 $n < 7$ 时, $y[n] = 0$, 当 $n > 23$ 时, $y[n] = 0$.

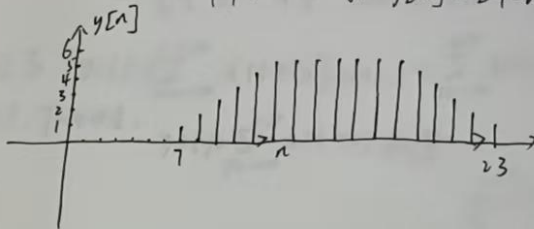
7A n	7	8	9	10	11	12	13	14	15	16	17	18	19
$y[n]$	1	2	3	4	5	6	6	6	6	6	6	6	5
n	20	21	22	23									
$y[n]$	4	3	2	1									

当 $7 \leq n \leq 11$ 时, $y[n] = n - 6$

当 $12 \leq n \leq 18$ 时, $y[n] = 6$.

当 $19 \leq n \leq 23$ 时, $y[n] = 24 - n$.

$$\Rightarrow y[n] = \begin{cases} n-6, & 7 \leq n \leq 11 \\ 6, & 12 \leq n \leq 18 \\ 24-n, & 19 \leq n \leq 23 \\ 0, & \text{其他} \end{cases}$$



2.11.

1a). $y(t) = \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau$, 由 $x(t) = 15h(t)$, $t \leq 3$ 时, $y(t) = 0$.

当 $3 < t \leq 5$ 时, $y(t) = \int_{-\infty}^{+\infty} u(\tau-3) e^{-3(t-\tau)} u(t-\tau) d\tau = \frac{1-e^{-3(t-3)}}{3}$

当 $5 < t \leq \infty$ 时, $y(t) = \frac{(1-e^{-6})e^{-3(t-5)}}{3}$

1b). $\frac{dy(t)}{dt} = \delta(t-3) - \delta(t-5)$ $\therefore g(t) = e^{-3(t-3)} u(t-3) - e^{-3(t-5)} u(t-5)$

1c). $g(t) = \frac{dy(t)}{dt}$

联系方式: _____

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2-17.

(a). 对 $\frac{dy(t)}{dt} + 4y(t) = x(t)$. 齐次解: $y_h(t) = Ce^{-4t}$.特解: 设 $y_p(t) = Ae^{(-1+j)t}$, 得 $A = \frac{1-j}{6}$. 3. 完全解: $y(t) = Ce^{-4t} + \frac{1-j}{6}e^{(-1+j)t}$
 $y(0^+) = 0 \Rightarrow C + \frac{1-j}{6} = 0, C = -\frac{1-j}{6}$

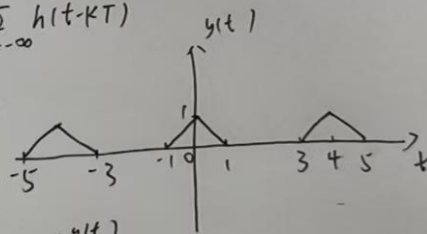
$$\therefore y(t) = \frac{1-j}{6} [e^{(-1+j)t} - e^{-4t}] u(t).$$

$$(b). y(t) = \operatorname{Re} \left\{ \frac{1-j}{6} [e^{(-1+j)t} - e^{-4t}] \right\} u(t) \neq 0$$

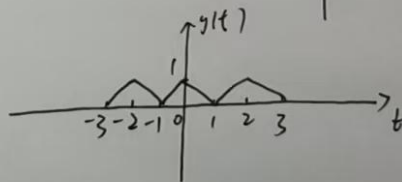
$$= \frac{1}{6} [-e^{-4t} + e^{-t} (\cos(3t) + \sin(3t))] u(t)$$

$$2-23 \quad y(t) = \left[\sum_{k=-\infty}^{+\infty} \delta(t-kT) \right] * h(t) = \sum_{k=-\infty}^{+\infty} h(t-kT)$$

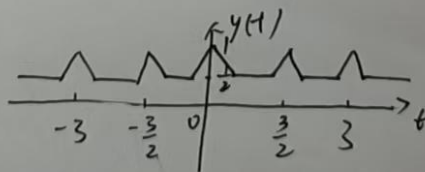
$$(a). T=4 \text{ 时}, y(t) = \sum_{k=-\infty}^{+\infty} h(t-4k) \text{ 如图:}$$



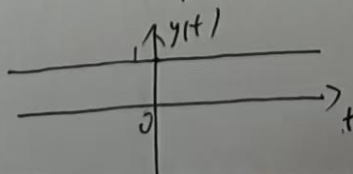
$$(b). y(t) = \sum_{k=-\infty}^{+\infty} h(t-2k)$$



$$(c). y(t) = \sum_{k=-\infty}^{+\infty} h(t - \frac{3}{2}k)$$



$$(d). y(t) = \sum_{k=-\infty}^{+\infty} h(t-k)$$



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